



Data Exploration: Visualizations



Department of
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Clemson® University



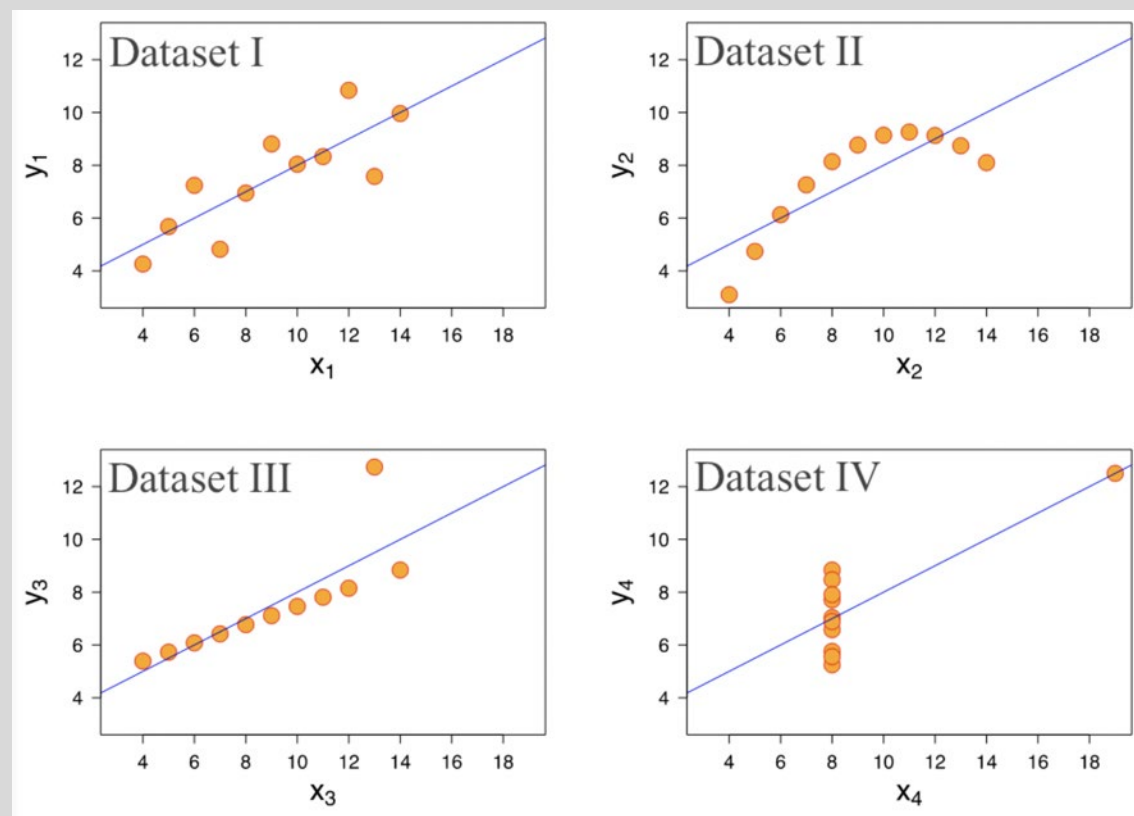
The following four data sets comprise the Anscombe's Quartet; all four sets of data have identical simple summary statistics.

Dataset I			Dataset II		Dataset III		Dataset IV	
x	y		x	y	x	y	x	y
10	8.04		10	9.14	10	7.46	8	6.58
8	6.95		8	8.14	8	6.77	8	5.76
13	7.58		13	8.74	13	12.74	8	7.71
9	8.81		9	8.77	9	7.11	8	8.84
11	8.33		11	9.26	11	7.81	8	8.47
14	9.96		14	8.1	14	8.84	8	7.04
6	7.24		6	6.13	6	6.08	8	5.25
4	4.26		4	3.1	4	5.39	19	12.5
12	10.84		12	9.13	12	8.15	8	5.56
7	4.82		7	7.26	7	6.42	8	7.91
5	5.68		5	4.74	5	5.73	8	6.89
Sum:	99.00	82.51	99.00	82.51	99.00	82.51	99.00	82.51
Avg:	9.00	7.50	9.00	7.50	9.00	7.50	9.00	7.50
Std:	3.32	2.03	3.32	2.03	3.32	2.03	3.32	2.03



Anscombe's Data

Summary statistics clearly don't tell the story of how they differ. But a picture can be worth a thousand words:

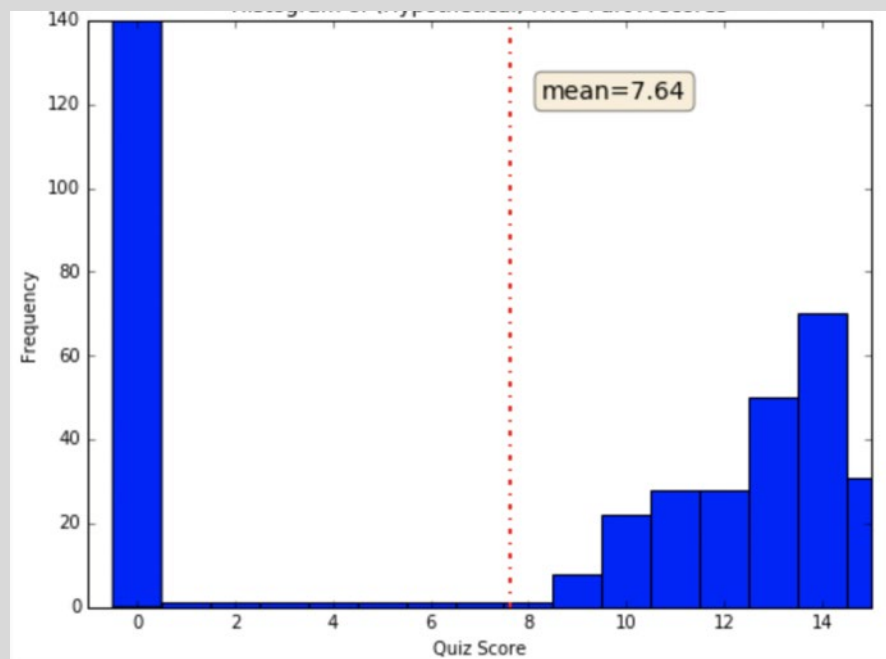




More Visualization Motivation

If I tell you that the average score for Homework 0 was: $7.64/15 = \underline{50.9\%}$ last year, what does that suggest?

And what does the graph suggest?





More Visualization Motivation

Visualizations help us to analyze and explore the data. They help to:

- Identify hidden patterns and trends
- Formulate/test hypotheses
- Communicate any modeling results
 - Present information and ideas succinctly
 - Provide evidence and support
 - Influence and persuade
- Determine the next step in analysis/modeling



Types of Visualizations

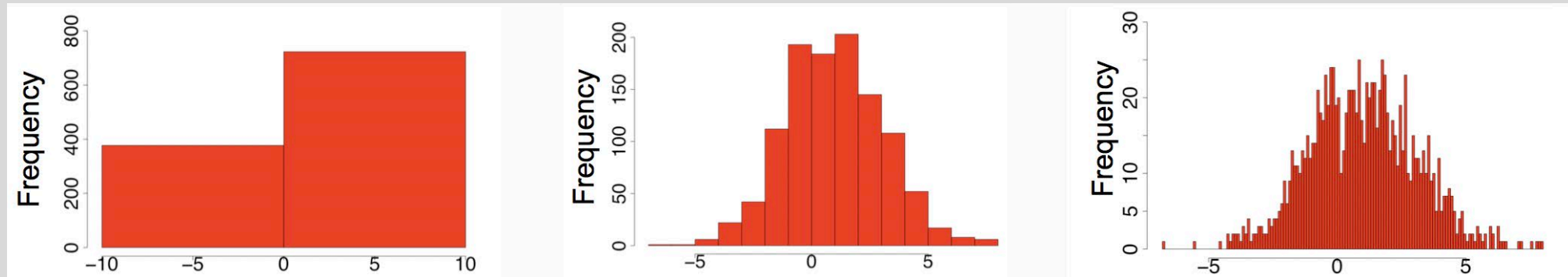
What do you want your visualization to show about your data?

- **Distribution:** how a variable or variables in the dataset distribute over a range of possible values.
- **Relationship:** how the values of multiple variables in the dataset relate
- **Composition:** how the dataset breaks down into subgroups
- **Comparison:** how trends in multiple variable or datasets compare



Histograms to Visualize Distribution

A **histogram** is a way to visualize how 1-dimensional data is distributed across certain values.



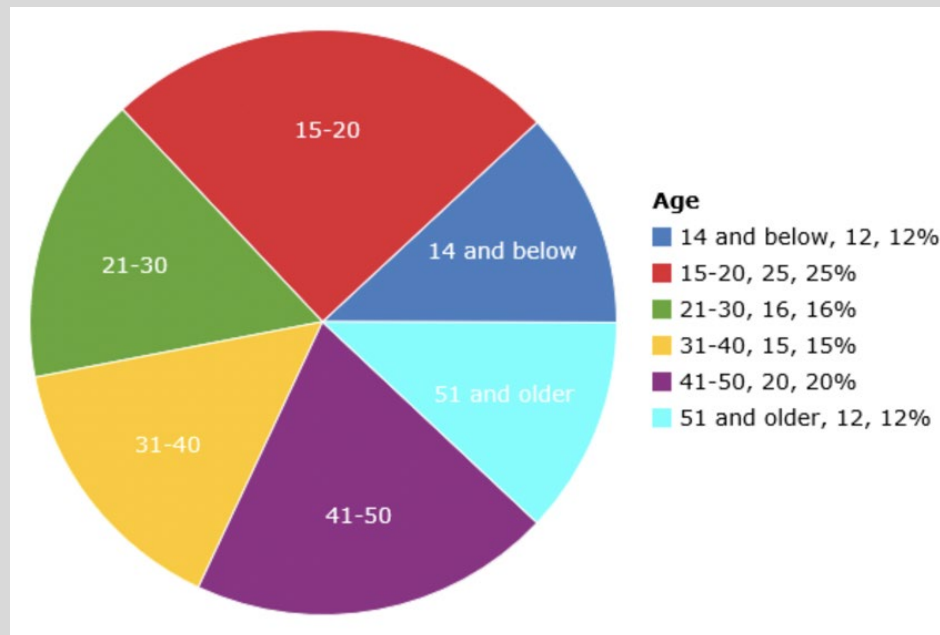
Note: Trends in histograms are sensitive to number of bins.



Pie Chart for a Categorical Variable?

A **pie chart** is a way to visualize the static composition (aka, distribution) of a variable (or single group).

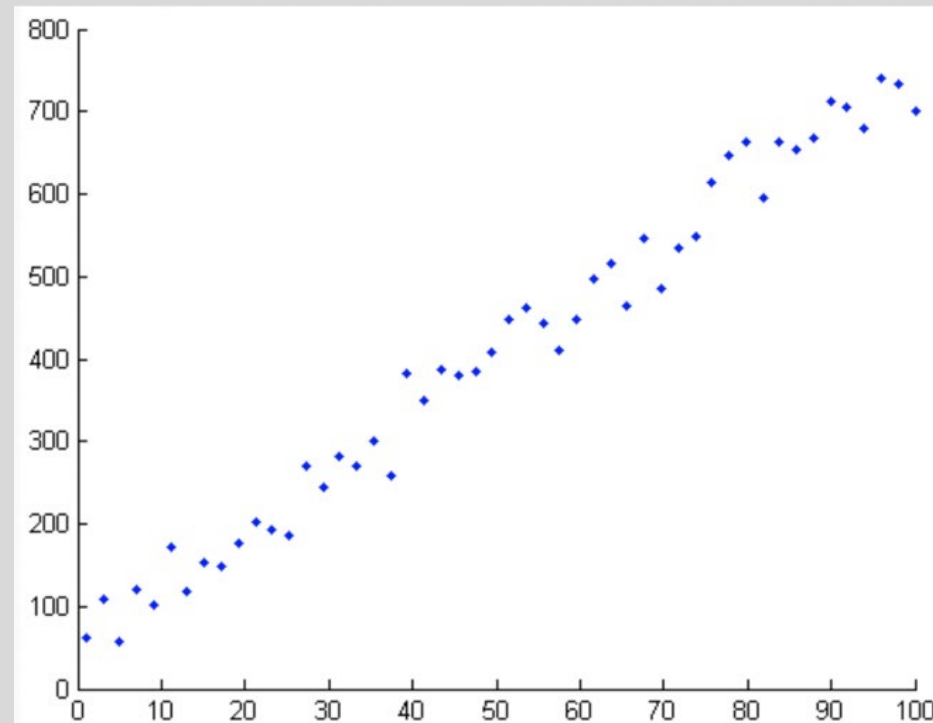
Pie charts are often frowned upon (and bar charts are used instead). Why?



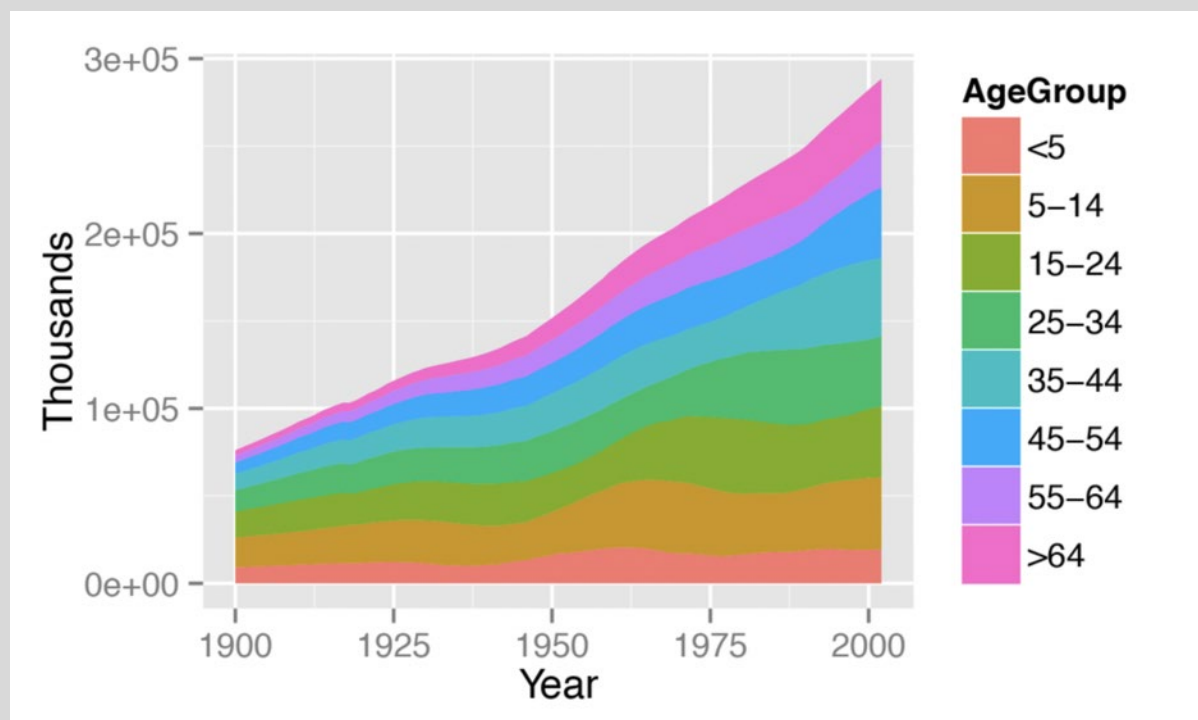


Scatter Plots to Visualize Relationships

A **scatter plot** is a way to visualize the relationship between two different attributes of multi-dimensional data.



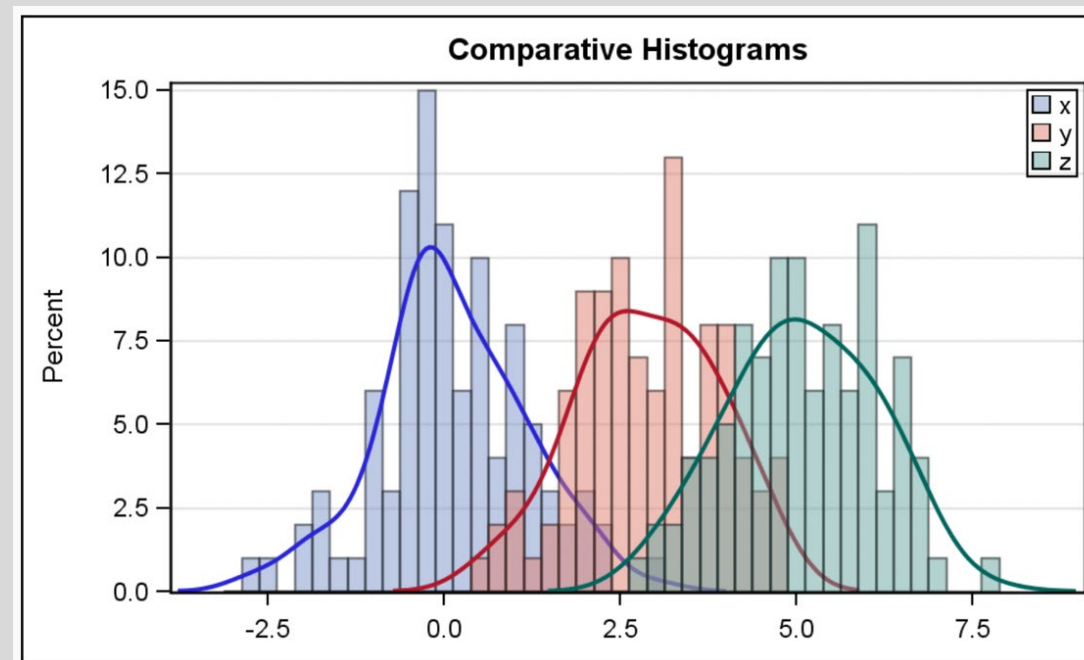
A **stacked area graph** is a way to visualize the composition of a group as it changes over time (or some other quantitative variable). This shows the relationship of a categorical variable (AgeGroup) to a quantitative variable (year).





Multiple Histograms

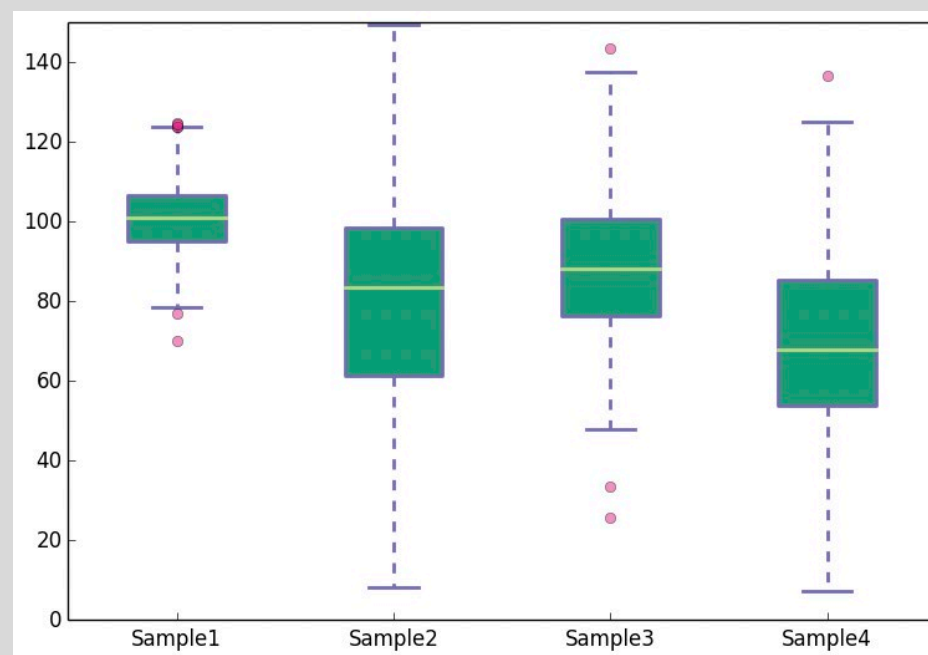
Plotting **multiple histograms** (and **kernel density estimates** of the distribution, here) on the same axes is a way to visualize how different variables compare (or how a variable differs over specific groups).





Boxplots

A **boxplot** is a simplified visualization to compare a quantitative variable across groups. It highlights the range, quartiles, median and any outliers present in a data set.





[Not] Anything is possible!

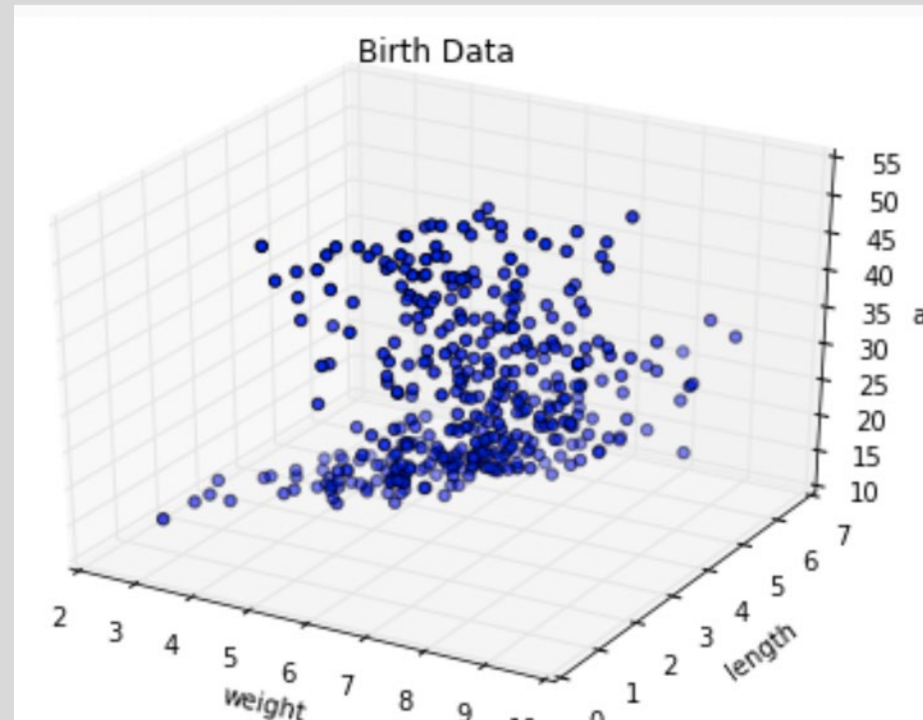
Often your dataset seem too complex to visualize:

- Data is too high dimensional (how do you plot 100 variables on the same set of axes?)
- Some variables are categorical (how do you plot values like Cat or No?)



More dimensions is not always better

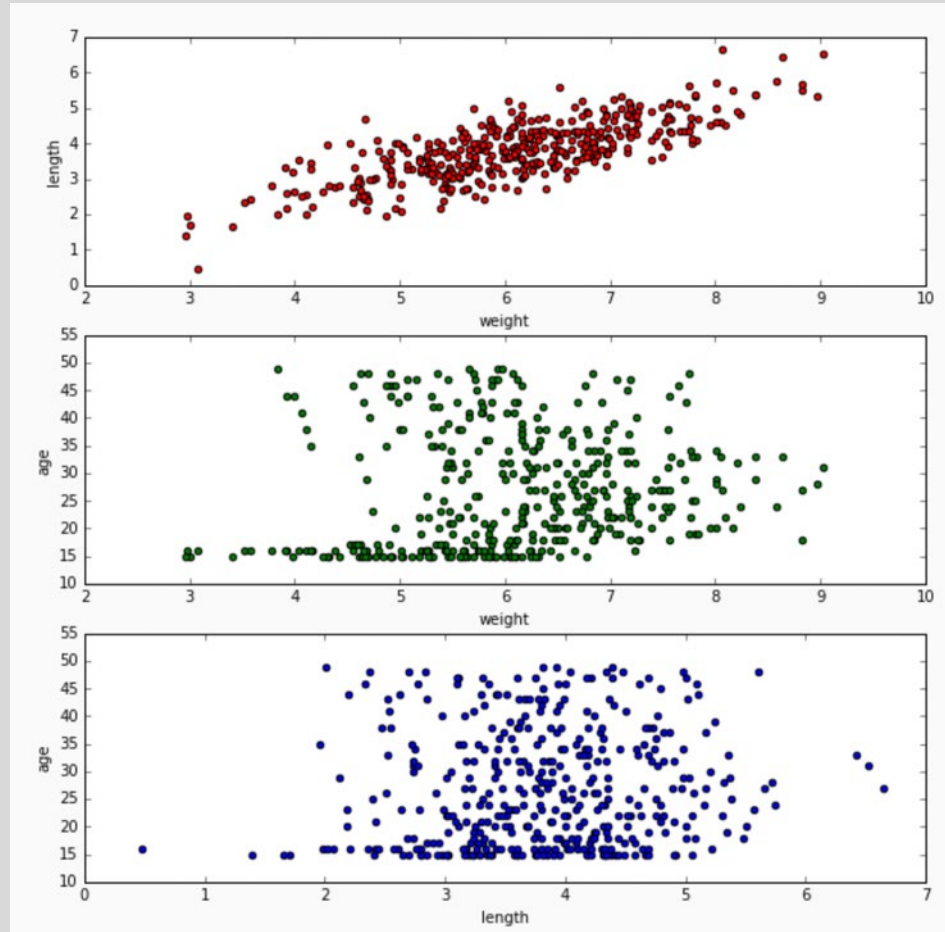
When the data is high dimensional, a scatter plot of all data attributes can be impossible or unhelpful





Reducing Complexity

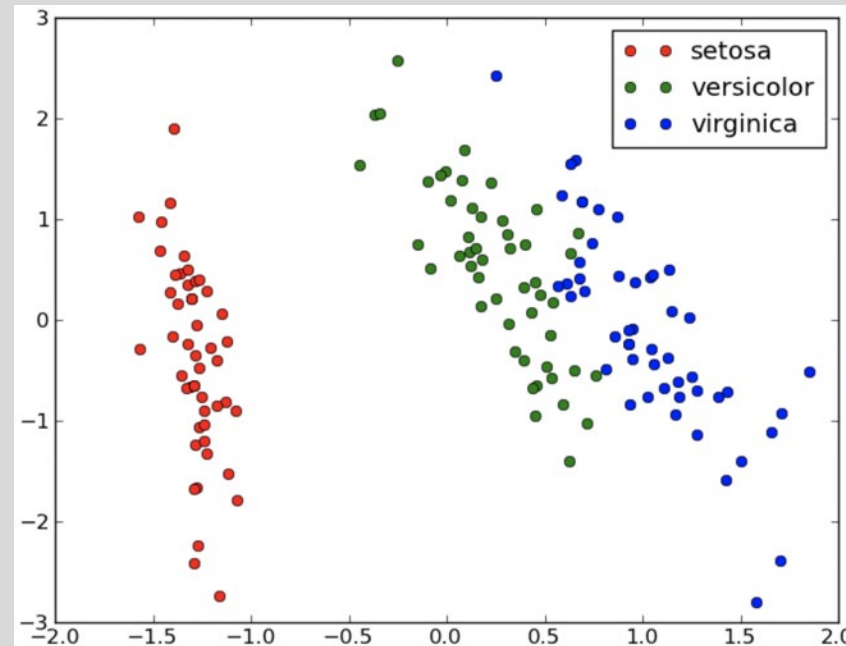
Relationships may be easier to spot by producing multiple plots of lower dimensionality.





Reducing Complexity

For 3D data, color coding a categorical attribute can be “effective.”



Except when it's not effective. What could be a better choice?



3D Can Work

For 3D data, a quantitative attribute can be encoded by size in a bubble chart.

The above visualizes a set of consumer products. The variables are: revenue, consumer rating, product type and product cost.

